Lecture 10

Mon 7/20 AM

Outline/topics

- EM basics
  - Maxwell's equations + boundary conditions
- Waveguide modes
- Dispersion relation
- Cavities
  - General
  - Accelerator-specific
- Important parameters

E&M review

- Maxwell's equations:

$$\vec{\nabla}\cdot\vec{D} = \rho \qquad \vec{\nabla}\cdot\vec{B} = 0$$

$$\vec{\nabla}\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t} \qquad \vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

where $\vec{D} = \varepsilon\vec{E}$, $\vec{B} = \mu\vec{H}$ $\begin{pmatrix}\varepsilon = \varepsilon_r\varepsilon_0 \\ \mu = \mu_r\mu_0\end{pmatrix}$

- EM waves/RF cavities:

time dependence $e^{i\omega t}$ $\omega \rightarrow$ frequency

Amperes + Faraday's laws become

$$\vec{\nabla}\times\vec{H} = \vec{J} + i\omega\vec{D} \qquad \vec{\nabla}\times\vec{E} = -i\omega\vec{B}$$

- Source free vacuum:

$\vec{J} = 0$, $\rho = 0$, $\vec{D} = \varepsilon_0\vec{E}$, $\vec{B} = \mu_0\vec{H}$

$\Rightarrow \vec{\nabla}\cdot\vec{E} = 0$, $\vec{\nabla}\cdot\vec{B} = 0$

$\vec{\nabla}\times\vec{H} = i\omega\varepsilon_0\vec{E}$, $\vec{\nabla}\times\vec{E} = -i\omega\mu_0\vec{H}$

- curl of each: $\vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \nabla^2\vec{A}$

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{H}) = i\omega\varepsilon_0\vec{\nabla}\times\vec{E} =$$

$$\vec{\nabla}(\overset{0}{\cancel{\vec{\nabla}\cdot\vec{H}}}) \cancel{-} \nabla^2\vec{H} = i\omega\varepsilon_0(\cancel{-}i\omega\mu_0\vec{H})$$

$$\nabla^2\vec{H} = \varepsilon_0\mu_0(i\omega)(i\omega)\vec{H} = -\varepsilon_0\mu_0\omega^2\frac{\partial^2\vec{H}}{\partial t^2}$$

* wave equation! $\nabla^2 f = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}$

- For $\vec{E}$: $\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -i\omega\mu_0\nabla\times\vec{H}$

$$\vec{\nabla}\cdot(\vec{\nabla}\cdot\vec{E}) \cancel{-} \nabla^2\vec{E} = \cancel{-}i\omega\mu_0(i\omega\varepsilon_0\vec{E})$$

$$\Rightarrow \boxed{\nabla^2\vec{E} = -\varepsilon_0\mu_0\omega^2\frac{\partial^2\vec{E}}{\partial t^2}}$$

- phase velocity $v_p = \frac{1}{\sqrt{\varepsilon_0\mu_0}} = c$ in vacuum
- In free space: plane waves

$$\vec{E} \perp \vec{B} \perp \vec{k} \leftarrow \text{wavevector}$$

wave number $k = \frac{2\pi}{\lambda} = \frac{\omega}{v}$

speed of wave, here c
but not always!

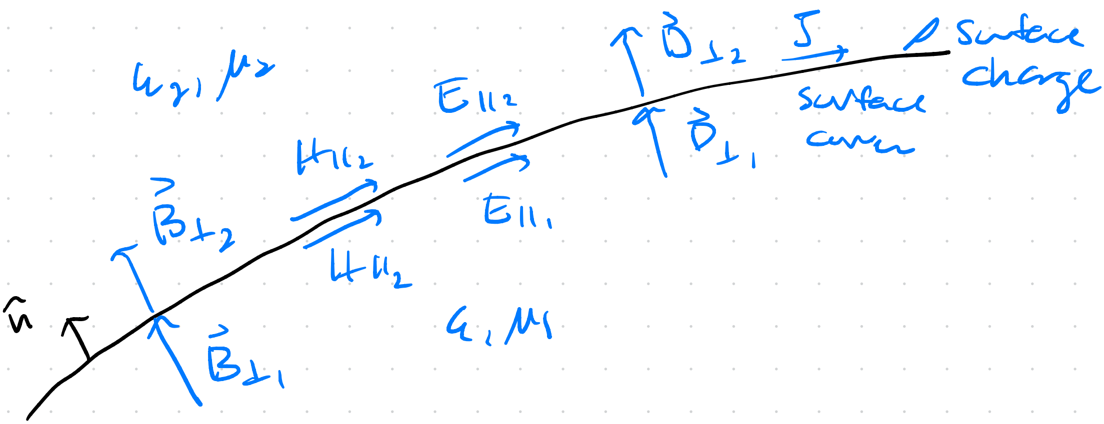
$$\nabla^2\vec{E} + k^2\vec{E} = 0 \qquad k^2 = \varepsilon_r\mu_r\frac{\omega^2}{c^2}$$

$$\nabla^2\vec{H} + k^2\vec{H} = 0$$

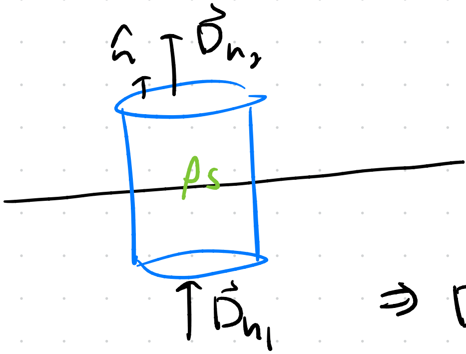
* error in H.W. eq 18.4 should be k^2

Boundary conditions

At interface between two media, restrictions on normal and tangential components of the field for continuity



- Arise from Maxwell's equations:



$$\oint_S \vec{D} \cdot d\vec{A} = \int_V \rho \, dV \qquad \vec{\nabla} \cdot \vec{D} = \rho$$

Gauss's law

$\Rightarrow D_{2n} - D_{1n} = \rho_s$ surface charge

$\Rightarrow \hat{n} \cdot (\vec{D}_2 - \vec{D}_1) = \rho_s$

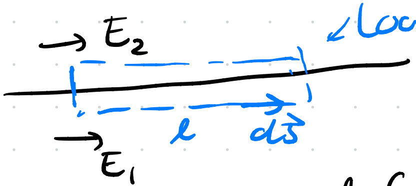
No magnetic charge $\Rightarrow \vec{B}_1 \cdot \hat{n} = \vec{B}_2 \cdot \hat{n}$

$\vec{\nabla} \cdot \vec{B} = 0$

- Conditions on E tangential to surface come from Faraday's law

$$\oint_C \vec{E}\cdot d\vec{s} = -\frac{d}{dt}\int_S \vec{B}\cdot d\vec{A}$$

loop to integrate



$$\vec{E}_1\cdot\vec{\ell} - \vec{E}_2\cdot\vec{\ell} = -\frac{d}{dt}\int_S \vec{B}\cdot d\vec{A}$$

As loop shrinks, flux vanishes

$$\Rightarrow \vec{E}_{1\parallel} - \vec{E}_{2\parallel} = 0$$

$\Rightarrow$ Field must be continuous across boundary to satisfy Ampere's law

- Conditions on $\vec{H}_\parallel$, same idea + Ampere's law:

$$\vec{H}_1\cdot\vec{\ell} - \vec{H}_2\cdot\vec{\ell} = I_{enc}$$

$$\Rightarrow \vec{H}_{1\parallel} - \vec{H}_{2\parallel} = \vec{J}_{surf}$$

↳ discontinuity in field produces a surface current

Conducting (metallic boundary)

- Fields cannot penetrate any substantial distance inside the bulk *   * skin depth
  - ↳ can prove from plane wave incident on boundary, $\sigma \to \infty$ total reflection

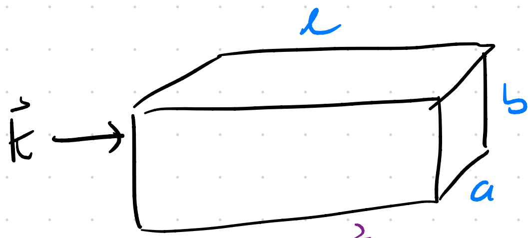
$$\Rightarrow \begin{aligned} &\hat{n}\cdot\vec{D} = \rho_s && \text{surface charge,} \\ &\hat{n}\times\vec{E} = 0 && \text{no tangential } \vec{E} \\ &\hat{n}\cdot\vec{B} = 0 && \text{no } \perp \vec{B} \text{ (or } \vec{H}) \\ &\hat{n}\times\vec{H} = \vec{J}_s && \text{tangential } \vec{H} \text{ results in } \vec{J}_s \end{aligned}$$

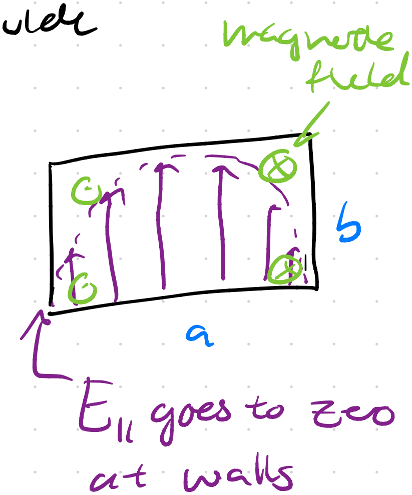
- Sets limitations on the shape of supported fields, giving rise to <u>eigenmodes</u>

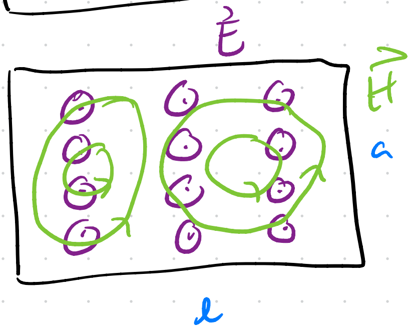
Guided waves

H.W. 18.1 ✶ slide

· Metallic rectangular waveguide







Changes:

like plane wave

· No longer purely transverse! Now can have some E_z, H_z

↳ introduces new requirements in solution to Maxwell's equations

· Waves only supported above a "cutoff" frequency. wave number:

$$k_z^2 = k^2 - k_c^2$$ ← cutoff

where $k_c^2 = k_x^2 + k_y^2$

$k_z \Rightarrow$ real for propagating wave

$$k_c^2 = k_x^2 + k_y^2$$

Cut-off frequency:

$$\omega_c = \frac{k_c}{c}$$

- Fields have the form (same for $\vec{H}$)

$$\vec{E}(x,y,z,t) = \vec{E}_0(x,y)\, e^{i(\omega t - k_z z)}$$

time evolution (ωt) propagation ($k_z z$)

$$\vec{E}_0 = E_x \hat{x} + E_y \hat{y} + E_z \hat{z}$$

* Apply Ampere's law + Faraday's law to find solutions:

$$\vec{\nabla} \times \vec{E} = -i\omega\mu_0 \vec{H} \qquad \vec{\nabla} \times \vec{H} = i\omega\epsilon_0 \vec{E}$$

$\Rightarrow$ 6 equations relating components

e.g. $$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -i\omega\mu H_z$$

$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = -i\omega\mu H_x \ \ldots$$

see all in e.g. Griffiths EM or by doing cross-products

- For a given configuration, we can write transverse terms (E_x, E_y, H_x, H_y) in terms of longitudinal $\Rightarrow$ 4 equations

* In lossy media, propagation is described by complex propagation constant

conductivity

$$\gamma = \alpha + i\beta = i\omega\left[\mu\varepsilon\left(1 - i\frac{\sigma}{\omega\varepsilon}\right)\right]^{1/2}$$

attenuation constant

phase constant

Naming convention: * slide

- TEM modes: evenly spaced with no cut-off frequency
  - ↳ i.e. can carry DC signals
- TM/TE eigenmodes have ω_c

Dispersion relation + velocities:

- Different in guided wave than in free space

$$v_{ph} = \frac{\omega}{k_z} \neq c$$

· Look at dispersion relation:

$$k_c = \frac{\omega_c}{c} \quad \text{and} \quad k_z^2 = k^2 - k_c^2$$

$$\boxed{k_z^2 = \frac{\omega^2}{v_{ph}^2} = \frac{\omega^2}{c^2} - \frac{\omega_c^2}{c^2}}$$ * slide

$$\Rightarrow \quad v_{ph}^2 = c^2 \frac{\omega^2}{\omega^2 - \omega_c^2}$$ phase velocity

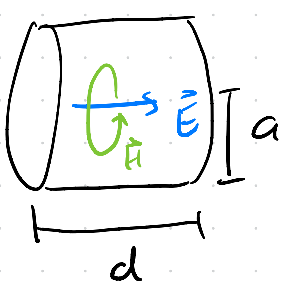
$$v_g = \frac{d\omega}{dk_z} = \frac{c^2 k_z}{\omega} < c$$ always group velocity

Resonant cavities

* see slides

Accelerator cavities

H.W. CH 18



ℓ = Length

a = radius

- Field seen by particle has the form

$$E_z = E_{z0} \sin\left(\omega \frac{z}{v} + \delta\right)$$

H.W. notation

cavity phase

- Energy gain

$$\Delta E_{kin} = |q| \int E_z \, dz$$

want particle to see correct sign of field throughout entire transit

- RF voltage defined as:

Assumes TM_{010}

$$V_{RF} = E_{z0} \, l = E_{010} \, l$$

accelerating gap

$$\Delta E_{gain} = |q| V_{RF} \frac{\sin(\omega d/2v)}{(\omega d/2v)} = |q| V_{cy}$$

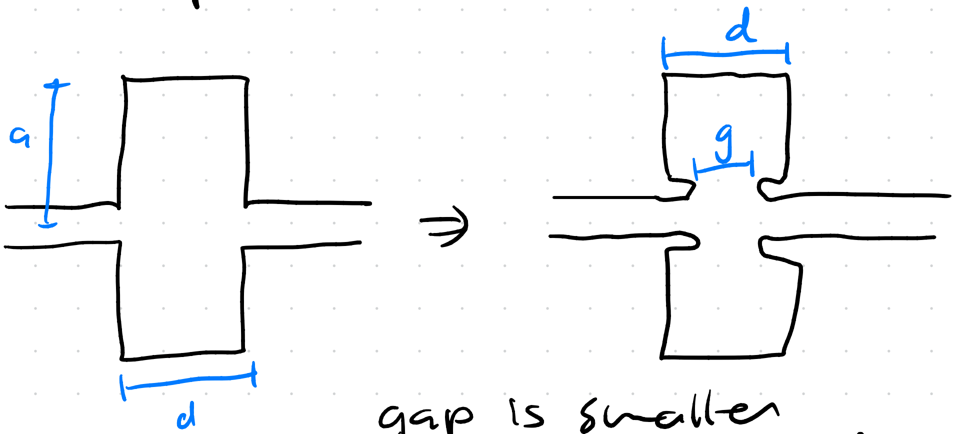
Effective cavity voltage, takes cycle into V

- Transit time factor:

$$T = \frac{\sin(\omega d/2v)}{(\omega d/2v)} \Rightarrow \Delta E_{kin} = |q| V_{RF} T$$

- can increase $T \Rightarrow$ increase ΔE_{kin} by making gap smaller

Example: nose cones



gap is smaller for same TM_{010} frequency

* see slides

Important parameters + limitations

- Finite conductivity of walls + magnetic fields creating surface currents in walls $\Rightarrow$ power dissipation

Recall: $\vec{H}_{1\parallel} - \vec{H}_{2\parallel} = \vec{J}_{surf}$

↳ discontinuity in field produces a surface current

$\hat{n} \times \vec{H} = \vec{J}_s$ tangential $\vec{H}$ results in $\vec{J}_s$

- $R_{surf} = \dfrac{1}{\sigma \delta_s}$ σ = conductivity (of walls) δ_s = skin depth (of walls)

- Power lost through surface:

per cycle

$$\frac{dP_{cy}}{dS} = R_{surf} J_s^2$$

cavity inner surface

* or in slides

$$P = \frac{R_{surf}}{2} \int_S |H_t|^2 dS$$

- For TM_{010} pillbox cavity (in vacuum):

$$P_{cy} = \frac{1}{4}\varepsilon_0 \omega \delta_s E_{010}^2 \int_S J_1^2(2.405\, r/a)\, dS$$

Bessel function!

Integrate + sub $E_{010} = V_{rf}/d$

$$\left[\, P_{cy} = \frac{1}{2}\pi \varepsilon_0 \omega \delta_s V_{RF}^2 J_1^2(2.405)\, \frac{a(a+d)}{d^2} \,\right]$$

* $J_1(2.405) = 0.51911\ldots$

* note H.W. uses μ for μ_r, ε for μ_r, always check your units!

- Energy stored is given by

For pillbox TM_{010}

$$U_e = \frac{\varepsilon}{4}\int_V \vec{E}\cdot\vec{E}^*\, dV \Rightarrow U_e = \frac{1}{2}\varepsilon_0 E_{010}^2\, d a^2 J_1^2(2.405)$$

- Q value: for a given mode

$$Q_0 = \frac{\omega U_e}{P_{cy}} \qquad \frac{\omega \cdot \text{energy stored}}{\text{power lost per cycle}}$$

(as in slides)

- Q_0 is <u>intrinsic</u> Q of the mode
- Q_L is achieved Q value when including coupling to mode

- Q_L determines "fill time" i.e. time needed to supply RF power before reaching full field

$$t_F = \frac{2Q_L}{\omega} = \frac{2Q_0}{(1+\beta_c)\omega}$$

coupling coefficient

Determines pulse length required from RF source

⇒ All impact pulsed heating ✱slides

- Shunt impedance: measure of achieved V_{cy} for some P_{cy}, want this to be high

$$R_s = \frac{V_{cy}^2}{2P_{cy}} \text{ measured in } \Omega$$

✱slides